

FitForge

Implemented Features

Your move, your progress.

This document lists every feature implemented in the FitForge system. Each section describes a major functional area, what it includes, and how it works. FitForge is an all-in-one gym management and workout tracking platform built with React, Node.js, MySQL, and MongoDB.

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FitForge Development Team

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FEATURE 1

Authentication & Accounts

A complete authentication system that supports three user roles and secure session management.

- User registration with email, password, optional username, full name, and birthday
- Login with email and password
- Password hashing using bcrypt (10 rounds)
- JWT authentication with 7-day token expiry
- Session persistence via localStorage and Axios interceptors
- Three user roles: user, coach, and admin
- Profile editing (name, username, and birthday)
- Automatic age calculation from birthday
- Auto-logout when the token expires or becomes invalid

FEATURE 2

Workout Tracking

Users can log every workout session with full detail on sets, reps, and weights. Personal records are detected automatically.

- Log a workout with name, date, notes, and unlimited sets
- Each set records exercise, weight, and reps
- Automatic personal record detection using the Epley 1RM formula
- Toast notification when new PRs are broken
- Workout history with expandable set details
- Delete workouts with cascade removal of all associated sets
- Activity logging — every workout is recorded in MongoDB

FEATURE 3

Routine Builder

Users can create reusable workout templates that pre-fill the workout logging form with one click.

- Create routines with name, description, and multiple exercises
- Each routine exercise has target sets, reps, and order index
- View routines with expandable exercise list
- Start a workout from a routine — pre-fills the form
- Delete routines with cascade removal of all associated exercises

FEATURE 4

Analytics Dashboard

A visual analytics page showing progress over time and strength balance across muscle groups.

- Volume over time — line chart of total kg lifted per day
- Muscle balance — pie chart showing volume distribution across muscle groups
- Personal records table — every PR with weight, muscle group, and date
- Range selector for 7, 30, 90, or 365-day windows

FEATURE 5

Exercise Library

A browsable, filterable library of 42+ exercises covering every major muscle group.

- 42 seeded exercises across six muscle groups
- Filter by muscle group (Chest, Back, Legs, Shoulders, Arms, Core)
- Filter by equipment type (Barbell, Dumbbell, Machine, Cable, Bodyweight)
- Search by exercise name
- Admin CRUD — add, edit, and delete exercises
- Deletion is blocked if the exercise is used in any workout, routine, or PR

FEATURE 6

Membership System

Three membership plans with full lifecycle management, expiry tracking, and automatic cancellation of previous memberships.

- Three plans: Walk-in (₱100/day), Monthly (₱1,200/month), Yearly (₱12,000/year)
- Auto-cancel previous active membership when subscribing to a new one
- Expiry tracking with days-remaining calculation
- Color-coded status: green (> 7 days), orange (≤ 7 days), red (expired)
- Active membership banner on the Profile page
- Full membership history on the Profile page
- Status transitions: Active, Cancelled, Cancelled + Refund
- Automatic payment refund when admin cancels with refund option

FEATURE 7

Payment Processing

A hybrid payment system supporting five methods, QR-code verification for digital payments, and full receipt management.

- Five payment methods: Cash, GCash, Maya, Bank Transfer, Card

- Instant activation for Cash and Card payments
- QR verification flow for GCash, Maya, and Bank Transfer:
 - Payment created with 'pending' status
 - QR code generated encoding a verification URL
 - Email sent with the QR code
 - User scans QR on phone and confirms payment
 - Admin verifies or rejects from the admin panel
 - Result email sent to the user
- Receipt generation for every transaction
- Receipt viewer accessible from Profile, Membership, Rentals, and Admin
- Send QR copy to up to 5 additional email addresses
- Resend QR email from QR modal, Profile, or Admin panel
- Optional auto-CC to admin billing address
- Pending payments queue in the admin panel with verify/reject actions
- Refund handling — payments flip to 'refunded' status on cancellation

FEATURE 8

Equipment Rentals

An hourly equipment rental system with live stock tracking, automatic receipts, and full admin control.

- 8 seeded rental items with hourly pricing
- Live stock tracking — decremented on reserve, restored on return
- Reserve equipment for a specific duration
- Return equipment with one click
- Status flow: Reserved → Confirmed → Hold → Active → Returned
- Automatic stock adjustment on admin status changes
- Admin CRUD for inventory — add, edit, and delete items
- Deletion blocked if active rentals reference an item
- Rental receipts emailed and viewable
- Admin status transitions with automatic stock adjustment and refunds

FEATURE 9

Personal Coaching

A coach-client system that lets gym coaches view their assigned clients' workout history and send detailed feedback.

- Coach role alongside user and admin
- Admin assigns clients to coaches via the Coaches tab
- Coach dashboard showing all assigned clients as cards

- Client detail view with stats, workout history, and recent PRs
- Send feedback with subject, content, and optional workout attachment
- Optional star rating (1 to 5 stars) on feedback
- Client 'My Coach' page showing all received feedback
- Unread badges on new feedback
- Mark as read — individual or bulk
- Coach can delete own feedback
- Dashboard stats — total clients, total feedback sent

FEATURE 10

Admin Dashboard

A comprehensive 10-tab dashboard giving administrators complete control over users, payments, memberships, rentals, coaching, inventory, and the exercise library.

- PROFILE — admin's own card plus grid of all users
- STATS — user counts, login chart, daily revenue, payment method pie, purpose breakdown
- USERS — full user table with role management
- ACTIVITY — MongoDB activity feed showing all user events
- PAYMENTS — all payments with revenue KPIs and receipt viewer
- PENDING — verification queue with Resend, Verify, Reject actions
- TRANSACTIONS — memberships and rentals tables with status dropdowns
- COACHES — assign coaches to clients, promote users, manage assignments
- INVENTORY — full CRUD for rental items
- EXERCISES — full CRUD for the exercise library

FEATURE 11

User Interface

A polished, responsive, dark-mode-friendly interface with smooth UX across all screens.

- Dark/light mode toggle
- Responsive layout for desktop, tablet, and mobile
- Toast notifications for success and error messages
- Loading states for all asynchronous operations
- Empty states with friendly messages when no data exists
- Modal components — QR display, receipt viewer, user detail drill-down
- Color-coded status pills throughout the app
- Confirmation prompts for destructive actions (delete, cancel, refund)
- Sticky navbar with role-aware navigation links
- Homepage with detailed About section

- Custom FitForge logo and favicon

FEATURE 12

Backend Systems

A robust backend built on a hybrid MySQL + MongoDB architecture with transactional guarantees and modular services.

- Hybrid database architecture — MySQL for relational data, MongoDB for logs and payments
- Transaction support for bulk inserts (workouts, rentals)
- Foreign key cascades for clean data deletion
- Activity logging service — every user action recorded in MongoDB
- Daily metrics rollup for fast admin dashboard queries
- Email service — Nodemailer with Gmail SMTP and HTML templates
- QR generation service — data URLs generated on demand
- PR detection service — Epley 1RM with strict-mode-safe CTE
- Recommendation service — suggests exercises for neglected muscle groups
- Centralized Express error handling middleware
- CORS support for local development
- Health check endpoint for monitoring

FEATURE 13

Data Model

A normalized, well-indexed schema designed for correctness and performance.

- 12 MySQL tables — users, exercises, routines, routine_exercises, workouts, workout_sets, personal_records, memberships, rental_items, rentals, coach_clients, coach_feedback
- 3 MongoDB collections — activity_logs, payment_logs, admin_metrics
- Seed data — 42 exercises, 8 rental items, 3 membership plans
- Unique constraints on email, username, and user+exercise PR pairs
- Indexed foreign keys for fast joins
- Composite indexes on frequently queried columns

End of Document

FitForge — Your move, your progress.